# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 83.8 ± 1.83 |
| Correlation Stability | 69.5 ± 4.15 |
| Distribution Consistency | 85.8 ± 1.40 |
| Entropy Preservation | 90.8 ± 0.75 |
| Mean Z Score Drift | 0.1 ± 0.01 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 23.0 ± 0.00 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 75.9 ± 0.30 |
| Feature Accuracy | 87.4 ± 1.85 |
| Novelty Score | 0.6 ± 0.02 |

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## Dataset Audit: heart_failure_clinical_records.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 299 |
| Columns | 13 |
| Numeric features | 13 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4.1-mini |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| age | 60.834 | 62.531 | +1.697 | 2.8% | OK |
| anaemia | 0.431 | 0.473 | +0.041 | 9.6% | OK |
| creatinine_phosphokinase | 581.839 | 345.285 | -236.555 | 40.7% | Drift |
| diabetes | 0.418 | 0.464 | +0.046 | 11.1% | Warn |
| ejection_fraction | 38.084 | 39.665 | +1.582 | 4.2% | OK |
| high_blood_pressure | 0.351 | 0.464 | +0.113 | 32.3% | Drift |
| platelets | 263358.029 | 253305.439 | -10052.590 | 3.8% | OK |
| serum_creatinine | 1.394 | 1.310 | -0.084 | 6.0% | OK |
| serum_sodium | 136.625 | 136.527 | -0.098 | 0.1% | OK |
| sex | 0.649 | 0.510 | -0.138 | 21.3% | Warn |
| smoking | 0.321 | 0.343 | +0.022 | 6.9% | OK |
| time | 130.261 | 135.527 | +5.266 | 4.0% | OK |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 83.8 ± 1.83 |
| Correlation Stability (*2) | 69.5 ± 4.15 |
| Distribution Consistency (*3) | 85.8 ± 1.40 |
| Entropy Preservation (*4) | 90.8 ± 0.75 |
| Mean Z Score Drift (*5) | 0.1 ± 0.01 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 23.0 ± 0.00 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 75.9 ± 0.30 |
| Feature Accuracy (*10) | 87.4 ± 1.85 |
| Novelty Score (*11) | 0.6 ± 0.02 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.4599999999999996 | 0.5283333333333333 | 0.5183333333333333 | 0.05833333333333335 | |
| F1 | 0.41719462248642464 | 0.18154403735799085 | 0.23881728979253358 | -0.17837733269389103 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6133333333333335 | 0.5666666666666667 | 0.5733333333333334 | -0.04 | |
| F1 | 0.007142857142857143 | 0.4443633440297351 | 0.2887289242626435 | 0.2815860671197863 | * |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.805 | 0.7666666666666667 | 0.7949999999999999 | -0.00999999999999998 | |
| F1 | 0.6894302795561282 | 0.6850420931549968 | 0.6950697151183475 | 0.005639435562219135 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7499999999999999 | 0.7300000000000001 | 0.7766666666666667 | 0.02666666666666666 | |
| F1 | 0.5667171896628354 | 0.6330829082428304 | 0.6739612984260884 | 0.10724410876325315 | * |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.785 | 0.67 | 0.7716666666666667 | -0.013333333333333341 | |
| F1 | 0.6833354404804183 | 0.5257700434068917 | 0.6750702940532054 | -0.008265146427212794 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6333333333333334 | 0.5700000000000001 | 0.6433333333333333 | 0.009999999999999986 | |
| F1 | N/A | 0.45012924671982635 | 0.20640957082133554 | 0.20640957082133554 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8316666666666667 | 0.7583333333333333 | 0.8116666666666668 | -0.019999999999999997 | |
| F1 | 0.7556715392060958 | 0.6457497835669488 | 0.7162128524887895 | -0.03945868671730628 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.835 | 0.7783333333333334 | 0.8266666666666668 | -0.008333333333333337 | |
| F1 | 0.7547418227770806 | 0.6595048426638571 | 0.746979452885834 | -0.007762369891246468 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.835 | 0.77 | 0.8483333333333333 | 0.013333333333333353 | |
| F1 | 0.7470875238282171 | 0.6365836824465857 | 0.769262444385174 | 0.022174920556956912 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.825 | 0.7683333333333333 | 0.8283333333333334 | 0.00333333333333332 14 | |
| F1 | 0.7417404771442427 | 0.654620548874084 | 0.7511140522208426 | 0.009373575076599782 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 2 | 55.0 | 0.0 | 7861.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.1 | 136.0 | 1.0 |
| 3 | 65.0 | 0.0 | 146.0 | 0.0 | 20.0 | 0.0 | 162000.0 | 1.3 | 129.0 | 1.0 |
| 4 | 50.0 | 1.0 | 111.0 | 0.0 | 20.0 | 0.0 | 210000.0 | 1.9 | 137.0 | 1.0 |
| 5 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 6 | 90.0 | 1.0 | 47.0 | 0.0 | 40.0 | 1.0 | 204000.0 | 2.1 | 132.0 | 1.0 |
| 7 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |
| 8 | 60.0 | 1.0 | 315.0 | 1.0 | 60.0 | 0.0 | 454000.0 | 1.1 | 131.0 | 1.0 |
| 9 | 65.0 | 0.0 | 157.0 | 0.0 | 65.0 | 0.0 | 263358.03 | 1.5 | 138.0 | 0.0 |
| 10 | 80.0 | 1.0 | 123.0 | 0.0 | 35.0 | 1.0 | 388000.0 | 9.4 | 133.0 | 1.0 |
| 11 | 75.0 | 1.0 | 81.0 | 0.0 | 38.0 | 1.0 | 368000.0 | 4.0 | 131.0 | 1.0 |
| 12 | 62.0 | 0.0 | 231.0 | 0.0 | 25.0 | 1.0 | 253000.0 | 0.9 | 140.0 | 1.0 |
| 13 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 14 | 50.0 | 1.0 | 168.0 | 0.0 | 38.0 | 1.0 | 276000.0 | 1.1 | 137.0 | 1.0 |
| 15 | 49.0 | 1.0 | 80.0 | 0.0 | 30.0 | 1.0 | 427000.0 | 1.0 | 138.0 | 0.0 |
| 16 | 82.0 | 1.0 | 379.0 | 0.0 | 50.0 | 0.0 | 47000.0 | 1.3 | 136.0 | 1.0 |
| 17 | 87.0 | 1.0 | 149.0 | 0.0 | 38.0 | 0.0 | 262000.0 | 0.9 | 140.0 | 1.0 |
| 18 | 45.0 | 0.0 | 582.0 | 0.0 | 14.0 | 0.0 | 166000.0 | 0.8 | 127.0 | 1.0 |
| 19 | 70.0 | 1.0 | 125.0 | 0.0 | 25.0 | 1.0 | 237000.0 | 1.0 | 140.0 | 0.0 |
| 20 | 48.0 | 1.0 | 582.0 | 1.0 | 55.0 | 0.0 | 87000.0 | 1.9 | 121.0 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 85.0 | 0.0 | 23.0 | 0.0 | 45.0 | 0.0 | 360000.0 | 3.0 | 132.0 | 1.0 |
| 2 | 55.0 | 0.0 | 748.0 | 0.0 | 45.0 | 0.0 | 263000.0 | 1.3 | 137.0 | 1.0 |
| 3 | 77.0 | 1.0 | 418.0 | 0.0 | 45.0 | 0.0 | 223000.0 | 1.8 | 145.0 | 1.0 |
| 4 | 60.0 | 1.0 | 231.0 | 1.0 | 25.0 | 0.0 | 194000.0 | 1.7 | 140.0 | 1.0 |
| 5 | 60.0 | 0.0 | 96.0 | 1.0 | 38.0 | 0.0 | 228000.0 | 0.75 | 140.0 | 0.0 |
| 6 | 70.0 | 0.0 | 835.0 | 0.0 | 35.0 | 1.0 | 305000.0 | 0.8 | 133.0 | 0.0 |
| 7 | 72.0 | 1.0 | 210.0 | 0.0 | 38.0 | 1.0 | 245000.0 | 1.2 | 134.0 | 1.0 |
| 8 | 59.0 | 0.0 | 85.0 | 1.0 | 45.0 | 0.0 | 310000.0 | 0.9 | 138.0 | 0.0 |
| 9 | 66.0 | 0.0 | 1200.0 | 0.0 | 30.0 | 1.0 | 180000.0 | 1.5 | 130.0 | 1.0 |
| 10 | 54.0 | 1.0 | 400.0 | 1.0 | 40.0 | 0.0 | 220000.0 | 1.0 | 140.0 | 0.0 |
| 11 | 61.0 | 0.0 | 300.0 | 0.0 | 50.0 | 0.0 | 270000.0 | 0.8 | 136.0 | 1.0 |
| 12 | 75.0 | 1.0 | 150.0 | 1.0 | 35.0 | 1.0 | 200000.0 | 1.8 | 132.0 | 0.0 |
| 13 | 45.0 | 1.0 | 66.0 | 1.0 | 25.0 | 0.0 | 233000.0 | 0.8 | 135.0 | 1.0 |
| 14 | 70.0 | 0.0 | 161.0 | 0.0 | 25.0 | 0.0 | 244000.0 | 1.2 | 142.0 | 0.0 |
| 15 | 58.0 | 1.0 | 400.0 | 0.0 | 40.0 | 0.0 | 164000.0 | 1.0 | 139.0 | 0.0 |
| 16 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 45.0 | 0.0 | 308.0 | 1.0 | 60.0 | 1.0 | 377000.0 | 1.0 | 136.0 | 1.0 |
| 2 | 60.0 | 1.0 | 2281.0 | 1.0 | 40.0 | 0.0 | 283000.0 | 1.0 | 141.0 | 0.0 |
| 3 | 55.0 | 0.0 | 582.0 | 1.0 | 35.0 | 1.0 | 371000.0 | 0.7 | 140.0 | 0.0 |
| 4 | 62.0 | 0.0 | 231.0 | 0.0 | 25.0 | 1.0 | 253000.0 | 0.9 | 140.0 | 1.0 |
| 5 | 50.0 | 0.0 | 369.0 | 1.0 | 25.0 | 0.0 | 252000.0 | 1.6 | 136.0 | 1.0 |
| 6 | 50.0 | 0.0 | 582.0 | 0.0 | 50.0 | 0.0 | 153000.0 | 0.6 | 134.0 | 0.0 |
| 7 | 67.0 | 0.0 | 410.0 | 1.0 | 40.0 | 1.0 | 280000.0 | 1.1 | 138.0 | 1.0 |
| 8 | 59.0 | 1.0 | 150.0 | 0.0 | 35.0 | 0.0 | 320000.0 | 0.9 | 140.0 | 0.0 |
| 9 | 72.0 | 0.0 | 85.0 | 1.0 | 50.0 | 1.0 | 180000.0 | 1.3 | 135.0 | 1.0 |
| 10 | 54.0 | 0.0 | 300.0 | 0.0 | 45.0 | 0.0 | 250000.0 | 0.8 | 137.0 | 0.0 |
| 11 | 63.0 | 1.0 | 600.0 | 1.0 | 25.0 | 1.0 | 210000.0 | 1.5 | 132.0 | 1.0 |
| 12 | 48.0 | 0.0 | 120.0 | 0.0 | 55.0 | 0.0 | 350000.0 | 0.7 | 139.0 | 0.0 |
| 13 | 60.0 | 0.0 | 2656.0 | 1.0 | 30.0 | 0.0 | 305000.0 | 2.3 | 137.0 | 1.0 |
| 14 | 45.0 | 1.0 | 130.0 | 0.0 | 35.0 | 0.0 | 174000.0 | 0.8 | 139.0 | 1.0 |
| 15 | 45.0 | 0.0 | 582.0 | 1.0 | 55.0 | 0.0 | 543000.0 | 1.0 | 132.0 | 0.0 |
| 16 | 65.0 | 1.0 | 59.0 | 1.0 | 60.0 | 0.0 | 172000.0 | 0.9 | 137.0 | 0.0 |

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## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 42.0 | 0.0 | 102.0 | 1.0 | 40.0 | 0.0 | 237000.0 | 1.2 | 140.0 | 1.0 |
| 2 | 85.0 | 0.0 | 5882.0 | 0.0 | 35.0 | 0.0 | 243000.0 | 1.0 | 132.0 | 1.0 |
| 3 | 65.0 | 0.0 | 892.0 | 1.0 | 35.0 | 0.0 | 263358.03 | 1.1 | 142.0 | 0.0 |
| 4 | 80.0 | 1.0 | 553.0 | 0.0 | 20.0 | 1.0 | 140000.0 | 4.4 | 133.0 | 1.0 |
| 5 | 58.0 | 0.0 | 144.0 | 1.0 | 38.0 | 1.0 | 327000.0 | 0.7 | 142.0 | 0.0 |
| 6 | 86.0 | 0.0 | 582.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 7 | 68.0 | 0.0 | 210.0 | 1.0 | 38.0 | 1.0 | 280000.0 | 1.1 | 138.0 | 1.0 |
| 8 | 75.0 | 1.0 | 450.0 | 0.0 | 25.0 | 1.0 | 190000.0 | 1.9 | 134.0 | 0.0 |
| 9 | 55.0 | 0.0 | 85.0 | 0.0 | 45.0 | 0.0 | 320000.0 | 0.8 | 140.0 | 1.0 |
| 10 | 62.0 | 1.0 | 600.0 | 1.0 | 30.0 | 0.0 | 230000.0 | 1.3 | 136.0 | 0.0 |
| 11 | 49.0 | 0.0 | 120.0 | 0.0 | 50.0 | 0.0 | 400000.0 | 0.7 | 142.0 | 1.0 |
| 12 | 80.0 | 1.0 | 700.0 | 0.0 | 20.0 | 1.0 | 150000.0 | 2.5 | 130.0 | 0.0 |
| 13 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |
| 14 | 79.0 | 1.0 | 55.0 | 0.0 | 50.0 | 1.0 | 172000.0 | 1.8 | 133.0 | 1.0 |
| 15 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 16 | 70.0 | 0.0 | 92.0 | 0.0 | 60.0 | 1.0 | 317000.0 | 0.8 | 140.0 | 0.0 |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.0 | 1.0 | 1082.0 | 1.0 | 45.0 | 0.0 | 250000.0 | 6.1 | 131.0 | 1.0 |
| 2 | 53.0 | 0.0 | 196.0 | 0.0 | 60.0 | 0.0 | 220000.0 | 0.7 | 133.0 | 1.0 |
| 3 | 55.0 | 0.0 | 582.0 | 1.0 | 35.0 | 1.0 | 371000.0 | 0.7 | 140.0 | 0.0 |
| 4 | 40.0 | 1.0 | 129.0 | 0.0 | 35.0 | 0.0 | 255000.0 | 0.9 | 137.0 | 1.0 |
| 5 | 45.0 | 1.0 | 1876.0 | 1.0 | 35.0 | 0.0 | 226000.0 | 0.9 | 138.0 | 1.0 |
| 6 | 58.0 | 1.0 | 57.0 | 0.0 | 25.0 | 0.0 | 189000.0 | 1.3 | 132.0 | 1.0 |
| 7 | 68.0 | 1.0 | 450.0 | 0.0 | 40.0 | 1.0 | 280000.0 | 1.2 | 138.0 | 1.0 |
| 8 | 55.0 | 0.0 | 1200.0 | 1.0 | 35.0 | 0.0 | 320000.0 | 0.8 | 140.0 | 0.0 |
| 9 | 72.0 | 1.0 | 85.0 | 0.0 | 25.0 | 1.0 | 180000.0 | 1.5 | 130.0 | 1.0 |
| 10 | 60.0 | 0.0 | 300.0 | 0.0 | 50.0 | 0.0 | 350000.0 | 0.7 | 136.0 | 0.0 |
| 11 | 65.0 | 1.0 | 75.0 | 1.0 | 30.0 | 1.0 | 210000.0 | 1.0 | 134.0 | 1.0 |
| 12 | 58.0 | 0.0 | 600.0 | 0.0 | 45.0 | 0.0 | 400000.0 | 0.9 | 139.0 | 0.0 |
| 13 | 48.0 | 1.0 | 131.0 | 1.0 | 30.0 | 1.0 | 244000.0 | 1.6 | 130.0 | 0.0 |
| 14 | 95.0 | 1.0 | 371.0 | 0.0 | 30.0 | 0.0 | 461000.0 | 2.0 | 132.0 | 1.0 |
| 15 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 16 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |

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## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 65.0 | 0.0 | 892.0 | 1.0 | 35.0 | 0.0 | 263358.03 | 1.1 | 142.0 | 0.0 |
| 2 | 55.0 | 0.0 | 748.0 | 0.0 | 45.0 | 0.0 | 263000.0 | 1.3 | 137.0 | 1.0 |
| 3 | 65.0 | 0.0 | 118.0 | 0.0 | 50.0 | 0.0 | 194000.0 | 1.1 | 145.0 | 1.0 |
| 4 | 60.0 | 1.0 | 260.0 | 1.0 | 38.0 | 0.0 | 255000.0 | 2.2 | 132.0 | 0.0 |
| 5 | 50.0 | 0.0 | 1548.0 | 0.0 | 30.0 | 1.0 | 211000.0 | 0.8 | 138.0 | 1.0 |
| 6 | 53.0 | 0.0 | 63.0 | 1.0 | 60.0 | 0.0 | 368000.0 | 0.8 | 135.0 | 1.0 |
| 7 | 67.0 | 0.0 | 450.0 | 1.0 | 40.0 | 1.0 | 210000.0 | 1.2 | 138.0 | 1.0 |
| 8 | 59.0 | 1.0 | 320.0 | 0.0 | 35.0 | 0.0 | 280000.0 | 1.0 | 140.0 | 0.0 |
| 9 | 72.0 | 0.0 | 1500.0 | 1.0 | 30.0 | 1.0 | 190000.0 | 1.5 | 135.0 | 1.0 |
| 10 | 54.0 | 0.0 | 85.0 | 0.0 | 50.0 | 0.0 | 350000.0 | 0.8 | 142.0 | 0.0 |
| 11 | 63.0 | 1.0 | 600.0 | 1.0 | 25.0 | 1.0 | 230000.0 | 2.0 | 130.0 | 1.0 |
| 12 | 58.0 | 0.0 | 400.0 | 0.0 | 45.0 | 0.0 | 270000.0 | 1.1 | 139.0 | 1.0 |
| 13 | 78.0 | 0.0 | 224.0 | 0.0 | 50.0 | 0.0 | 481000.0 | 1.4 | 138.0 | 1.0 |
| 14 | 70.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 1.0 |
| 15 | 61.0 | 0.0 | 248.0 | 0.0 | 30.0 | 1.0 | 267000.0 | 0.7 | 136.0 | 1.0 |
| 16 | 40.0 | 0.0 | 478.0 | 1.0 | 30.0 | 0.0 | 303000.0 | 0.9 | 136.0 | 1.0 |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 60.0 | 1.0 | 754.0 | 1.0 | 40.0 | 1.0 | 328000.0 | 1.2 | 126.0 | 1.0 |
| 2 | 51.0 | 0.0 | 1380.0 | 0.0 | 25.0 | 1.0 | 271000.0 | 0.9 | 130.0 | 1.0 |
| 3 | 45.0 | 0.0 | 582.0 | 0.0 | 14.0 | 0.0 | 166000.0 | 0.8 | 127.0 | 1.0 |
| 4 | 85.0 | 0.0 | 129.0 | 0.0 | 60.0 | 0.0 | 306000.0 | 1.2 | 132.0 | 1.0 |
| 5 | 47.0 | 0.0 | 582.0 | 0.0 | 25.0 | 0.0 | 130000.0 | 0.8 | 134.0 | 1.0 |
| 6 | 70.0 | 0.0 | 97.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 138.0 | 1.0 |
| 7 | 67.0 | 0.0 | 150.0 | 1.0 | 35.0 | 1.0 | 230000.0 | 1.1 | 138.0 | 1.0 |
| 8 | 58.0 | 1.0 | 400.0 | 0.0 | 40.0 | 0.0 | 280000.0 | 0.9 | 135.0 | 0.0 |
| 9 | 72.0 | 0.0 | 85.0 | 1.0 | 50.0 | 1.0 | 210000.0 | 1.3 | 140.0 | 1.0 |
| 10 | 49.0 | 0.0 | 300.0 | 0.0 | 25.0 | 0.0 | 310000.0 | 0.8 | 132.0 | 0.0 |
| 11 | 63.0 | 1.0 | 1200.0 | 1.0 | 30.0 | 1.0 | 180000.0 | 1.5 | 134.0 | 1.0 |
| 12 | 55.0 | 0.0 | 95.0 | 0.0 | 45.0 | 0.0 | 250000.0 | 0.7 | 136.0 | 0.0 |
| 13 | 81.0 | 0.0 | 4540.0 | 0.0 | 35.0 | 0.0 | 231000.0 | 1.18 | 137.0 | 1.0 |
| 14 | 70.0 | 0.0 | 2695.0 | 1.0 | 40.0 | 0.0 | 241000.0 | 1.0 | 137.0 | 1.0 |
| 15 | 70.0 | 0.0 | 835.0 | 0.0 | 35.0 | 1.0 | 305000.0 | 0.8 | 133.0 | 0.0 |
| 16 | 80.0 | 0.0 | 776.0 | 1.0 | 38.0 | 1.0 | 192000.0 | 1.3 | 135.0 | 0.0 |

# TabuLLM: AI Benchmark Report

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 80.0 | 0.0 | 582.0 | 1.0 | 35.0 | 0.0 | 350000.0 | 2.1 | 134.0 | 1.0 |
| 2 | 68.0 | 1.0 | 157.0 | 1.0 | 60.0 | 0.0 | 208000.0 | 1.0 | 140.0 | 0.0 |
| 3 | 55.0 | 1.0 | 170.0 | 1.0 | 40.0 | 0.0 | 336000.0 | 1.2 | 135.0 | 1.0 |
| 4 | 50.0 | 0.0 | 1548.0 | 0.0 | 30.0 | 1.0 | 211000.0 | 0.8 | 138.0 | 1.0 |
| 5 | 52.0 | 0.0 | 190.0 | 1.0 | 38.0 | 0.0 | 382000.0 | 1.0 | 140.0 | 1.0 |
| 6 | 70.0 | 0.0 | 582.0 | 0.0 | 40.0 | 0.0 | 51000.0 | 2.7 | 136.0 | 1.0 |
| 7 | 67.0 | 0.0 | 210.0 | 1.0 | 40.0 | 1.0 | 280000.0 | 1.1 | 137.0 | 1.0 |
| 8 | 58.0 | 1.0 | 95.0 | 0.0 | 55.0 | 0.0 | 320000.0 | 0.9 | 140.0 | 0.0 |
| 9 | 72.0 | 0.0 | 450.0 | 1.0 | 35.0 | 1.0 | 150000.0 | 1.5 | 133.0 | 1.0 |
| 10 | 49.0 | 0.0 | 130.0 | 0.0 | 60.0 | 0.0 | 270000.0 | 0.8 | 142.0 | 1.0 |
| 11 | 63.0 | 1.0 | 85.0 | 1.0 | 50.0 | 0.0 | 310000.0 | 1.0 | 139.0 | 0.0 |
| 12 | 55.0 | 0.0 | 600.0 | 0.0 | 30.0 | 1.0 | 200000.0 | 1.3 | 135.0 | 1.0 |
| 13 | 75.0 | 0.0 | 582.0 | 1.0 | 30.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 14 | 53.0 | 0.0 | 207.0 | 1.0 | 40.0 | 0.0 | 223000.0 | 1.2 | 130.0 | 0.0 |
| 15 | 60.0 | 1.0 | 2281.0 | 1.0 | 40.0 | 0.0 | 283000.0 | 1.0 | 141.0 | 0.0 |
| 16 | 59.0 | 1.0 | 280.0 | 1.0 | 25.0 | 1.0 | 302000.0 | 1.0 | 141.0 | 0.0 |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 41.0 | 0.0 | 148.0 | 0.0 | 40.0 | 0.0 | 374000.0 | 0.8 | 140.0 | 1.0 |
| 2 | 60.0 | 0.0 | 253.0 | 0.0 | 35.0 | 0.0 | 279000.0 | 1.7 | 140.0 | 1.0 |
| 3 | 50.0 | 1.0 | 121.0 | 1.0 | 40.0 | 0.0 | 260000.0 | 0.7 | 130.0 | 1.0 |
| 4 | 68.0 | 1.0 | 1021.0 | 1.0 | 35.0 | 0.0 | 271000.0 | 1.1 | 134.0 | 1.0 |
| 5 | 81.0 | 0.0 | 4540.0 | 0.0 | 35.0 | 0.0 | 231000.0 | 1.18 | 137.0 | 1.0 |
| 6 | 55.0 | 0.0 | 47.0 | 0.0 | 35.0 | 1.0 | 173000.0 | 1.1 | 137.0 | 1.0 |
| 7 | 67.0 | 0.0 | 210.0 | 1.0 | 35.0 | 1.0 | 280000.0 | 1.2 | 138.0 | 1.0 |
| 8 | 54.0 | 1.0 | 95.0 | 0.0 | 40.0 | 0.0 | 320000.0 | 0.9 | 140.0 | 0.0 |
| 9 | 72.0 | 0.0 | 580.0 | 1.0 | 25.0 | 1.0 | 250000.0 | 1.8 | 130.0 | 1.0 |
| 10 | 48.0 | 0.0 | 130.0 | 0.0 | 50.0 | 0.0 | 300000.0 | 0.7 | 142.0 | 0.0 |
| 11 | 80.0 | 1.0 | 400.0 | 1.0 | 30.0 | 1.0 | 270000.0 | 1.5 | 135.0 | 1.0 |
| 12 | 62.0 | 0.0 | 75.0 | 0.0 | 45.0 | 0.0 | 350000.0 | 0.8 | 140.0 | 1.0 |
| 13 | 65.0 | 0.0 | 224.0 | 1.0 | 50.0 | 0.0 | 149000.0 | 1.3 | 137.0 | 1.0 |
| 14 | 80.0 | 0.0 | 898.0 | 0.0 | 25.0 | 0.0 | 149000.0 | 1.1 | 144.0 | 1.0 |
| 15 | 66.0 | 1.0 | 72.0 | 0.0 | 40.0 | 1.0 | 242000.0 | 1.2 | 134.0 | 1.0 |
| 16 | 75.0 | 1.0 | 582.0 | 0.0 | 30.0 | 0.0 | 225000.0 | 1.83 | 134.0 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.0 | 0.0 | 92.0 | 0.0 | 60.0 | 1.0 | 317000.0 | 0.8 | 140.0 | 0.0 |
| 2 | 70.0 | 1.0 | 59.0 | 0.0 | 60.0 | 0.0 | 255000.0 | 1.1 | 136.0 | 0.0 |
| 3 | 63.0 | 1.0 | 103.0 | 1.0 | 35.0 | 0.0 | 179000.0 | 0.9 | 136.0 | 1.0 |
| 4 | 60.0 | 0.0 | 253.0 | 0.0 | 35.0 | 0.0 | 279000.0 | 1.7 | 140.0 | 1.0 |
| 5 | 65.0 | 0.0 | 56.0 | 0.0 | 25.0 | 0.0 | 237000.0 | 5.0 | 130.0 | 0.0 |
| 6 | 45.0 | 0.0 | 2060.0 | 1.0 | 60.0 | 0.0 | 742000.0 | 0.8 | 138.0 | 0.0 |
| 7 | 68.0 | 0.0 | 110.0 | 1.0 | 40.0 | 1.0 | 280000.0 | 1.2 | 137.0 | 1.0 |
| 8 | 74.0 | 1.0 | 85.0 | 0.0 | 30.0 | 0.0 | 230000.0 | 1.5 | 134.0 | 0.0 |
| 9 | 59.0 | 0.0 | 560.0 | 0.0 | 55.0 | 0.0 | 350000.0 | 0.7 | 140.0 | 1.0 |
| 10 | 62.0 | 1.0 | 300.0 | 1.0 | 25.0 | 1.0 | 270000.0 | 1.8 | 132.0 | 0.0 |
| 11 | 70.0 | 0.0 | 150.0 | 0.0 | 35.0 | 0.0 | 210000.0 | 1.0 | 136.0 | 1.0 |
| 12 | 65.0 | 0.0 | 90.0 | 0.0 | 50.0 | 0.0 | 320000.0 | 0.9 | 138.0 | 0.0 |
| 13 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |
| 14 | 60.0 | 0.0 | 68.0 | 0.0 | 20.0 | 0.0 | 119000.0 | 2.9 | 127.0 | 1.0 |
| 15 | 70.0 | 0.0 | 97.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 138.0 | 1.0 |
| 16 | 53.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 249000.0 | 0.7 | 138.0 | 1.0 |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 58.0 | 0.0 | 132.0 | 1.0 | 38.0 | 1.0 | 253000.0 | 1.0 | 139.0 | 1.0 |
| 2 | 70.0 | 0.0 | 582.0 | 0.0 | 40.0 | 0.0 | 51000.0 | 2.7 | 136.0 | 1.0 |
| 3 | 54.0 | 1.0 | 427.0 | 0.0 | 70.0 | 1.0 | 151000.0 | 9.0 | 137.0 | 0.0 |
| 4 | 53.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 249000.0 | 0.7 | 138.0 | 1.0 |
| 5 | 46.0 | 0.0 | 719.0 | 0.0 | 40.0 | 1.0 | 263358.03 | 1.18 | 137.0 | 0.0 |
| 6 | 90.0 | 1.0 | 60.0 | 1.0 | 50.0 | 0.0 | 226000.0 | 1.0 | 134.0 | 1.0 |
| 7 | 67.0 | 0.0 | 210.0 | 1.0 | 40.0 | 1.0 | 280000.0 | 1.1 | 138.0 | 1.0 |
| 8 | 55.0 | 1.0 | 85.0 | 0.0 | 55.0 | 0.0 | 310000.0 | 0.8 | 140.0 | 0.0 |
| 9 | 72.0 | 0.0 | 1200.0 | 1.0 | 30.0 | 1.0 | 190000.0 | 1.5 | 135.0 | 1.0 |
| 10 | 48.0 | 0.0 | 300.0 | 0.0 | 60.0 | 0.0 | 350000.0 | 0.7 | 142.0 | 0.0 |
| 11 | 80.0 | 1.0 | 75.0 | 1.0 | 25.0 | 1.0 | 220000.0 | 2.0 | 130.0 | 1.0 |
| 12 | 62.0 | 0.0 | 400.0 | 0.0 | 50.0 | 0.0 | 270000.0 | 1.0 | 136.0 | 1.0 |
| 13 | 60.0 | 0.0 | 897.0 | 1.0 | 45.0 | 0.0 | 297000.0 | 1.0 | 133.0 | 1.0 |
| 14 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 15 | 45.0 | 1.0 | 1876.0 | 1.0 | 35.0 | 0.0 | 226000.0 | 0.9 | 138.0 | 1.0 |
| 16 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |